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Image processing apparatus and control method therefor

Abstract

An image processing apparatus includes an acquisition unit for acquiring an image obtained by an image capturing apparatus including an image sensor for capturing a light beam from a subject through an optical lens, and a generation unit for generating first subject distance information indicating a distance from the image capturing apparatus to the subject based on information corresponding to an entrance pupil position of the optical lens used to capture the image, a focus position, and second subject distance information indicating a distance from the image sensor to the subject.

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Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to combining processing in visual effects (VFX).

Description of the Related Art

(2) In the field of video production, visual effects (VFX) are used to provide special screen effects

(visual effects) to create non-realistic images and are realized by combining computer graphics (CG) with captured real images. The combining processing is typically performed at a post-production stage after images are captured. Recently, there has been a growing need for creating CG in advance and capturing images while checking a video image combined with the CG in real time on site. In the combining processing, it is important to set imaging ranges such that the imaging range of captured real images in image capturing matches the imaging range of CG. Accordingly, the combining processing is performed using imaging parameters (hereinafter referred to as metadata) superimposed on each frame in the imaging range of the captured real video image. Japanese Patent Application Laid-Open No. 2020-10269 discusses a technique in which a subject distance is defined as the total value of a focal length from a front-side principal point position of an imaging lens and a distance from a focal position to a subject, and a change in magnification of an image along with the movement of a focus lens is corrected. In addition, Japanese Patent Application Laid-Open No. 2011-78008 discusses a technique for reliably identifying a subject in each content, easily managing camerawork information and the like, and acquiring content data in consideration of the relationships among materials of various elements when content data is created using combinations of materials.
(3) The subject distance discussed in Japanese Patent Application Laid-Open No. 2020-10269 is a distance from an entrance pupil position within a lens to an in-focus subject. This subject distance is different from a distance from an imaging plane to a subject that is normally recognized as the subject distance by a user, and cannot be used for combining an image with computer graphics (CG). In addition, the subject distance discussed in Japanese Patent Application Laid-Open No. 2011-78008 is a distance measured by an automatic focusing mechanism included in an optical focus adjustment unit. This subject distance cannot be used to combine an image with CG, accordingly.
SUMMARY
(4) The present disclosure is directed to providing an image capturing apparatus and an image processing apparatus that make it possible to perform CG combining processing even when the user cannot recognize the subject distance that is used for CG combining processing and is defined as the distance from the imaging plane to the subject.
(5) According to an aspect of the present disclosure, an image processing apparatus includes an acquisition unit configured to acquire an image obtained by an image capturing apparatus including an image sensor configured to capture a light beam from a subject through an optical lens, and a generation unit configured to generate first subject distance information indicating a distance from the image capturing apparatus to the subject based on information corresponding to an entrance pupil position of the optical lens used to capture the image, a focus position, and second subject distance information indicating a distance from the image sensor to the subject.
(6) According to another aspect of the present disclosure, a control method for an image processing apparatus includes acquiring an image obtained by an image capturing apparatus including an image sensor configured to capture a light beam from a subject through an optical lens, and generating first subject distance information indicating a distance from the image capturing apparatus to the subject based on information corresponding to an entrance pupil position of the optical lens used to capture the image, a focus position, and second subject distance information indicating a distance from the image sensor to the subject.
(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram illustrating a configuration example of an image capturing apparatus according to a first exemplary embodiment.

(2) FIG. **2** illustrates relationships among optical distances according to first, second, third, and fourth exemplary embodiments.

(3) FIG. **3** is a flowchart illustrating computer graphics (CG) combining metadata generation processing according to the first exemplary embodiment.

(4) FIG. **4** is a flowchart illustrating CG combining metadata generation processing according to the first exemplary embodiment.

(5) FIG. **5** is a block diagram illustrating a configuration example of an image processing apparatus according to the second exemplary embodiment.

(6) FIG. **6** is a flowchart illustrating CG combining metadata generation processing according to the second exemplary embodiment.

(7) FIG. **7** is a flowchart illustrating CG combining processing using metadata according to the second exemplary embodiment.

(8) FIG. **8** is a flowchart illustrating CG combining metadata generation processing according to the third exemplary embodiment.

(9) FIG. **9** is a flowchart illustrating CG combining metadata generation processing according to the fourth exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

(10) Exemplary embodiments of the present disclosure will now be described with reference to the drawings. The following exemplary embodiments are not meant to limit the scope of the present disclosure as encompassed by the claims. While multiple features are described in the following exemplary embodiments, not all of the features described in the exemplary embodiments are necessarily deemed to be essential for the present disclosure and the features may be combined as appropriate.

First Exemplary Embodiment

(11) FIG. **1** is a block diagram illustrating an image capturing apparatus **100** according to a first exemplary embodiment. The image capturing apparatus **100** includes a lens **101**, a lens control unit **102**, a mechanical shutter **103**, and a shutter control unit **104**. The image capturing apparatus **100** further includes an image sensor **105**, an image signal processing circuit **106**, a memory unit **107**, a system control unit **108**, a system memory **109**, a display unit **110**, and a recording medium interface (I/F) unit **111**. The image capturing apparatus **100** still further includes a recording medium **112**, an external I/F unit **113**, an operation unit **114**, a lens information acquisition unit **115**, a parameter storage unit **116**, an imaging information generation unit **117**, and an imaging information output unit **118**.

(12) The lens **101** includes an optical lens and various mechanisms for implementing, for example, a zooming mechanism, a focusing mechanism, a diaphragm mechanism, and an image shake correction mechanism. The optical lens and the various mechanisms are driven and controlled by the lens control unit **102** or a focus ring and a zoom lens that are mounted on the lens **101**. A light beam from a subject passes through the lens **101** and the light beam is focused on an imaging plane on the image sensor **105**, and thereby a subject image is captured. The lens **101** can be a lens integrated with the image capturing apparatus **100**, or can be an interchangeable lens formed separately from the image capturing apparatus **100**. The subject image formed on the imaging plane of the image sensor **105** is photoelectrically converted. Further, a gain adjustment and an analog-to-digital (A/D) conversion to convert an analog signal into a digital signal are performed on the subject image. The digital signal thus obtained is transmitted to the image signal processing circuit **106** as R, Gr, Gb, and B signals. The image signal processing circuit **106** performs various image signal processing, such as development processing, low-pass filter processing to reduce noise,

shading processing, white balance (WB) processing, and cyclic noise reduction (NR) processing, and further performs various correction processing, image data compression processing, and the like. The mechanical shutter **103** is driven and controlled by the shutter control unit **104**. The memory unit **107** temporarily stores image data.

(13) The system control unit **108** performs the overall control operation and various arithmetic operations on the image capturing apparatus **100**. The system memory **109** stores arithmetic processing results from the system control unit **108**. The display unit **110** displays image data. The recording medium I/F unit **111** records image data into the recording medium **112** or reads out image data from the recording medium **112**. The external I/F unit **113** is an interface for communicating with an external computer, a video image input device or the like. The operation unit **114** is used for the user to input drive conditions and to make various settings for the image capturing apparatus **100**. The drive conditions and various setting values are transmitted to the system control unit **108**. The system control unit **108** performs the overall control operation for the image capturing apparatus **100** based on these pieces of information.

(14) The lens information acquisition unit **115** acquires a focal length, a focus position, a subject distance, a field angle, and the like corresponding to the driving status of the lens **101**, and sends the acquired lens information to the system control unit **108**. The parameter storage unit **116** stores parameters to be used for the imaging information generation unit **117** to generate information, provides the parameters to the system control unit **108**, and holds parameters input by the operation unit **114**.

(15) The imaging information generation unit **117** generates imaging information based on information about, for example, the lens **101**, the image sensor **105**, the lens information acquisition unit **115**, and the parameter storage unit **116**, each of which constituting the image capturing apparatus **100**. The generated imaging information and information about the image capturing apparatus **100** are stored in a predetermined area as metadata to be associated with image data.

(16) FIG. **2** illustrates an example of imaging information generated by the imaging information generation unit **117**. The imaging information generation unit **117** generates at least one of information about a horizontal field angle and a vertical field angle and CG combining subject distance information illustrated in FIG. **2**. The imaging information output unit **118** outputs various information generated by the imaging information generation unit **117** and information acquired by the lens information acquisition unit **115** via the external I/F unit **113** or the recording medium I/F unit **111**.

(17) An operation of the imaging information generation unit **117** according to the present exemplary embodiment having the configuration as described above will now be described with reference to the block diagram illustrated in FIG. **1** and flowcharts illustrated in FIGS. **3** and **4**. The processing illustrated in FIGS. **3** and **4** is repeatedly executed on an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle is used for the predetermined cycle.

(18) In step S**301**, the system control unit **108** determines imaging conditions and sets an exposure area for the image sensor **105**. In the present exemplary embodiment, a super 35 mm area is set for a 35 mm fullsize sensor, but instead any other imaging range can be set. In step S**302**, the system control unit **108** sets an imaging position, a focal length, a focus position, and the like at initial positions. These settings can be made by using the focus ring or the zoom ring mounted on the lens **101**, or through communication from the operation unit **114** to the lens **101**.

(19) In step S**303**, the system control unit **108** starts video image recording. Depending on mode settings, the system control unit **108** can start video image delivering via the external I/F unit **113** without recording the video image. In this case, the focal length, the focus position, and the like can be changed at any time.

(20) In step S**304**, the system control unit **108** acquires lens information, including subject distance

information (first subject distance information) and field angles (first horizontal field angle and first vertical field angle), from the lens information acquisition unit **115**.

(21) In step S**305**, the system control unit **108** determines whether the lens information acquired in step S**304** includes entrance pupil distance information as information corresponding to the entrance pupil position.

(22) The entrance pupil distance information to be acquired is information that is preliminarily stored in a memory in the lens **101** or a server, is received by the image capturing apparatus **100** at an arbitrary timing, and is stored in the memory unit **107** in the image capturing apparatus **100**. Alternatively, the entrance pupil distance information can be acquired during the operation in step S**304**. If the system control unit **108** determines that the acquired lens information includes the entrance pupil distance information (YES in step S**305**), the processing proceeds to step S**306**. If the system control unit **108** determines that the acquired lens information does not include the entrance pupil distance information (NO in step S**305**), the processing proceeds to step S**307**.

(23) In step S**306**, the system control unit **108** generates second subject distance information based on optical information including the first subject distance information and the entrance pupil distance information acquired in step S**304**. Then, the processing proceeds to step S**309**. In this case, the second subject distance information corresponds to an appropriate subject distance to be used for CG combining processing. As illustrated in FIG. **2**, when the first subject distance information (the subject distance) is represented by OD1, the entrance pupil distance is represented by EP, and the second subject distance information (CG combining subject distance) is represented by OD2, the following formula (1) can be obtained.

$$OD2=OD1-EP \quad (1)$$

(24) In step S**307**, the system control unit **108** acquires a coefficient in place of the entrance pupil distance information as information corresponding to the entrance pupil position. The coefficient acquired in place of the entrance pupil distance information is obtained by, for example, calculating the entrance pupil distance based on the actual measured value of the subject distance in each state of the lens **101**, calculating a coefficient for converting the entrance pupil distance to a subject distance for CG combining processing, and storing the calculated coefficient. The coefficient acquired in place of the entrance pupil distance information can be acquired from the parameter storage unit **116**. More alternatively, an input value obtained by using the operation unit **114** can be used as the coefficient, or the coefficient can be acquired via the external I/F unit **113**. In step S**308**, third subject distance information is generated based on the coefficient acquired in place of the entrance pupil distance information acquired in step S**307** and the first subject distance information acquired in step S**304**. Then, the processing proceeds to step S**309**. When the first subject distance information is represented by OD1, the coefficient acquired in place of the entrance pupil distance information is represented by K1, and the third subject distance information is represented by OD3, the following formula (2) can be obtained.

$$OD3=OD1-K1 \quad (2)$$

(25) In step S**309**, the system control unit **108** generates image capturing information, such as a second horizontal field angle and a second vertical field angle, and selects the second subject distance information or the third subject distance information as CG combining subject distance information. When the first horizontal field angle is represented by AOV1_H, the first vertical field angle is represented by AOV1_V, the second horizontal field angle is represented by AOV2_H, the second vertical field angle is represented by AOV2_V, the diagonal length of the 35 mm fullsize sensor is represented by 35 mmFullSize_HV, and the diagonal length of the super 35 mm sensor is represented by Super35 mm_HV, the following formulas (3) and (4) can be obtained.

(26) $$AOV2\_H = AOV1\_H \text{ .Math. } \frac{Super35mm\_HV}{35mmFullSize\_HV} \quad (3)$$

$$AOV2\_V = AOV1\_V \text{ .Math. } \frac{Super35mm\_HV}{35mmFullSize\_HV} \quad (4)$$

(27) In step S**310**, the system control unit **108** associates the imaging information generated in step

S**309** and the pieces of lens information acquired in step S**304** with frames obtained during image capturing, and stores the imaging information and lens information in association with the frames in the system memory **109**. For example, the imaging information and lens information can be associated with the frames by synchronizing the frames with the various information using a time code. If video image recording is started in step S**303**, the recording can be completed in step S**311**.
(28) An operation of the imaging information output unit **118** according to the present exemplary embodiment having the configuration as described above will now be described with reference to the block diagram illustrated in FIG. **1** and the flowchart illustrated in FIG. **4**. The processing illustrated in FIG. **4** is repeatedly executed on an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle is used for the predetermined cycle.
(29) First, in step S**401**, the system control unit **108** converts a data format for each mode for outputting various information generated by the imaging information generation unit **117** and information acquired by the lens information acquisition unit **115** (the converted data format being hereinafter referred to as metadata). In the present exemplary embodiment, a data format for outputting data via a network or a serial data interface (SDI) via the external I/F unit **113** and a data format for outputting data to the recording medium **112** through the recording medium I/F unit **111** are used by way of example. In step S**402**, the type of metadata to be output is selected. As the metadata to be output, metadata to be used for CG combining processing may be extracted, or all metadata may be output. Alternatively, the metadata may be changed based on a processing load on the image capturing apparatus **100** or other restrictions. In step S**403**, if the video image and metadata are output in synchronization with the frames of the captured image in real time, the metadata is output via a network or an SDI through the external I/F unit **113**, and if the video image is recorded once, the metadata is output to the recording medium **112** through the recording medium I/F unit **111**. Alternatively, these processes can be combined at once. If the video image and metadata are output in real time, a CG combining processing apparatus can be connected to the external I/F unit **113** to perform CG combining processing in real time. The frames and metadata to be output in this case can also be delayed depending on the CG combining processing apparatus.
(30) As described above, in the present exemplary embodiment, even when the user cannot recognize information for CG combining processing, the information for CG combining processing is generated based on an entrance pupil position or information about a coefficient corresponding to the entrance pupil position, thereby making it possible to provide metadata for the user to accurately combine a video image with CG.

Second Exemplary Embodiment

(31) In the first exemplary embodiment described above, metadata for CG combining processing is generated using the image capturing apparatus **100** by way of example. A second exemplary embodiment illustrates an example where an image processing apparatus combines an image with CG using metadata output with an image captured by the image capturing apparatus **100** described above in the first exemplary embodiment. FIG. **5** is a block diagram illustrating a configuration example of an image processing apparatus **500**. The image processing apparatus **500** illustrated in FIG. **5** has a configuration different from the configuration according to the first exemplary embodiment, or is integrally formed with the image capturing apparatus **100** according to the first exemplary embodiment. The image processing apparatus **500** performs CG combining processing by superimposing CG and metadata synchronized with frames obtained from the image capturing apparatus **100** on the frames.
(32) A CG input unit **501** inputs CG to be superimposed on a CG combining image to a CG combining unit **505**.
(33) A video image input unit **502** inputs the CG combining image. In this case, the video image input unit **502** can be connected to the image capturing apparatus **100** and can input a captured video image in real time, or can input a video image recorded on the image capturing apparatus

100. A metadata acquisition unit **503** acquires CG combining metadata synchronized with the image. The CG combining metadata can be acquired from the metadata superimposed on the image input by the video image input unit **502**, or only the metadata can be acquired separately from the video image. More alternatively, the CG combining metadata can be acquired via a network, or can be directly input by the user. An imaging information generation unit **504** generates CG combining information (CG combining image field information) based on the metadata acquired by the metadata acquisition unit **503**.

(34) The CG combining unit **505** performs CG combining processing by superimposing the CG input by the CG input unit **501** on the image input by the video image input unit **502** using the information generated by the imaging information generation unit **504**. A combined image output unit **506** outputs the combined image obtained by the CG combining unit **505** to a display apparatus, a recording medium, an external output I/F, and the like. As an output form, the combined image can be recorded on another recording medium through the external I/F, or can be transmitted to another connected video image input device, or can be transmitted via a network. An operation unit **507** is used to input drive conditions and make various settings for the image processing apparatus **500**. The drive conditions and various setting values are transmitted to the imaging information generation unit **504** and the CG combining unit **505**. A control unit **510** performs the overall control operation for the image processing apparatus **500** based on these pieces of information.

(35) An operation of the imaging information generation unit **504** according to the present exemplary embodiment having the configuration as described above will now be described with reference to the block diagram illustrated in FIG. **5** and a flowchart illustrated in FIG. **6**. The processing illustrated in FIG. **6** is repeatedly executed at an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle for combining processing is used for the predetermined cycle.

(36) In step S**601**, the control unit **510** determines whether the metadata that is acquired by the metadata acquisition unit **503** and associated with the image includes entrance pupil position information. If it is determined that the metadata includes the entrance pupil position information (YES in step S**601**), the processing proceeds to step S**602**. If it is determined that the metadata does not include the entrance pupil position information (NO in step S**601**), the processing proceeds to step S**603**.

(37) In step S**602**, the control unit **510** generates the second subject distance information based on the information acquired by the metadata acquisition unit **503**, such as the first subject distance information, the entrance pupil distance information as information corresponding to the entrance pupil position, and the focus position. The processing then proceeds to step S**605**. As described above in the first exemplary embodiment, formula (1) can be obtained, where the first subject distance information is represented by OD1, the entrance pupil position is represented by EP, and the second subject distance information is represented by OD2.

(38) In step S**603**, the control unit **510** acquires a coefficient in place of the entrance pupil position information from the metadata acquisition unit **503**. The coefficient acquired in place of the entrance pupil position information is similar to that acquired in the first exemplary embodiment. The coefficient can be acquired from the metadata acquisition unit **503**, or can be acquired from the operation unit **507**.

(39) In step S**604**, the control unit **510** generates the third subject distance information based on the coefficient acquired in place of the entrance pupil position information in step S**603**, the lens information acquired by the metadata acquisition unit **503**, such as the first subject distance information and the entrance pupil position, imaging range information, and the like. The processing then proceeds to step S**605**. When the first subject distance information is represented by OD1, the coefficient acquired in place of the entrance pupil position information is represented by K1, and the third subject distance information is represented by OD3, formula (2) can be

obtained. In step S**605**, it is determined whether the metadata acquired by the metadata acquisition unit **503** includes first horizontal field angle information and first vertical field angle information. If it is determined that the acquired metadata includes the first horizontal field angle information and the first vertical field angle information (YES in step S**605**), the processing proceeds to step S**606**. If it is determined that the acquired metadata does not include the first horizontal field angle information and the first vertical field angle information, the processing proceeds to step S**607**.
(40) In step S**606**, the control unit **510** generates second horizontal field angle information and second vertical field angle information based on the information acquired by the metadata acquisition unit **503**, such as the first horizontal field angle information, the first vertical field angle information, the lens information, and the image capturing information. The processing then proceeds to step S**609**. When the first horizontal field angle is represented by AOV1_H, the first vertical field angle is represented by AOV1_V, the second horizontal field angle is represented by AOV2_H, the second vertical field angle is represented by AOV2_V, the diagonal length of the 35 mm fullsize sensor is represented by 35 mmFullSize_HV, and the diagonal length of the super 35 mm sensor is represented by Super35 mm_HV, formulas (3) and (4) can be obtained.
(41) In step S**607**, the control unit **510** acquires focal length information from the metadata acquisition unit **503**. The focal length information can be acquired from the metadata acquisition unit **503** or the operation unit **507**.
(42) In step S**608**, the control unit **510** generates third horizontal field angle information and third vertical field angle information based on the focal length information acquired in place of the first horizontal field angle information and the first vertical field angle information in step S**607**, the imaging range information acquired by the metadata acquisition unit **503**, and the like. Then, the processing proceeds to step S**609**.
(43) In step S**609**, the control unit **510** determines CG combining subject distance information based on the second subject distance information generated in step S**602** or the third subject distance information generated in step S**604**. In step S**610**, the control unit **510** determines CG combining field angle information based on the second horizontal field angle information and the second vertical field angle information generated in step S**606** or the third horizontal field angle information and the third vertical field angle information generated in step S**608**. In step S**611**, CG combining image field information is calculated based on the CG combining subject distance information determined in step S**609** and the CG combining field angle information determined in step S**610**.
(44) An operation of the CG combining unit **505** according to the present exemplary embodiment having the configuration as described above will now be described with reference to the block diagram illustrated in FIG. **5** and a flowchart illustrated in FIG. **7**. The processing illustrated in FIG. **7** is repeatedly executed at an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle for combining processing is used for the predetermined cycle. In step S**701**, an image used for CG combining processing, a CG object, a background to be combined, metadata information acquired by the metadata acquisition unit **503**, information generated by the imaging information generation unit **504**, and the like are prepared. These pieces of information are loaded to CG combining software.
(45) The CG object used for CG combining processing and the background to be combined can be input through the CG input unit **501**, or can be developed to be used in the software.
(46) In step S**702**, the control unit **510** corrects a distortion, marginal illumination, or the like on the image to be combined based on the metadata information acquired by the metadata acquisition unit **503** and the information generated by the imaging information generation unit **504**. In the present exemplary embodiment, the CG combining subject distance information and CG combining field angle information generated by the imaging information generation unit **504**, the CG combining image field information, and the like are used.
(47) In step S**703**, the control unit **510** calculates the positional relationship between the CG object

and the background to be combined with the image to be combined by using the metadata information acquired by the metadata acquisition unit **503** and the information generated by the imaging information generation unit **504**. The control unit **510** then determines the combining positions of the CG object and the background.

(48) In step S**704**, the CG combining unit **505** adds a depth-of-field blur and a distortion in the CG object and the background to be combined with the image to be combined by using the metadata information acquired by the metadata acquisition unit **503** and the information generated by the imaging information generation unit **504**.

(49) While the exemplary embodiments of the present disclosure have been described in detail above, the present disclosure is not limited to the specific exemplary embodiments. Various modifications can be made without departing from the scope of the disclosure.

(50) While the above-described exemplary embodiments illustrate an example where the subject distance information is generated based on the entrance pupil position acquired through communication from the lens **101** or acquired by inputting any value by the user, the subject distance information can be generated based on information acquired by any other method. For example, entrance pupil position data depending on the focal length and focus position of a lens to be used can be held in a camera and can be referenced as needed. Alternatively, the entrance pupil position depending on the focal length and focus position of a lens to be used can be referenced on a web via a network.

Third Exemplary Embodiment

(51) While the first exemplary embodiment described above illustrates an example where subject distance information is generated as metadata for CG combining processing in the image capturing apparatus **100**, a third exemplary embodiment illustrates an example where information about field angles is generated as metadata for CG combining processing in the image capturing apparatus **100**. An example of an imaging information generation method will be described with reference to the block diagram illustrated in FIG. **1** and a flowchart illustrated in FIG. **8**. The processing illustrated in FIG. **8** is repeatedly executed on an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle is used for the predetermined cycle.

(52) In step S**801**, the system control unit **108** determines imaging conditions and sets an exposure area for the image sensor **105**. In the present exemplary embodiment, a super 35 mm area is set in a 35 mm fullsize sensor, but instead any other imaging range can be set.

(53) In step S**802**, the system control unit **108** sets the imaging position, the focal length, the focus position, and the like at initial positions. These settings can be made using the focus ring or the zoom ring mounted on the lens **101**, or through communication from the operation unit **114** to the lens **101**.

(54) In step S**803**, the system control unit **108** can start video image recording, or video image delivering via the external I/F unit **113** without recording the video image. In this case, the focal length, the focus position, and the like can be changed at any time. In step S**804**, the system control unit **108** acquires lens information including focal length information and field angle information (first horizontal field angle and first vertical field angle) from the lens information acquisition unit **115**.

(55) In step S**805**, the system control unit **108** determines whether the lens information acquired in step S**804** includes the field angle information. If the system control unit **108** determines that the acquired lens information includes the field angle information (YES in step S**805**), the processing proceeds to step S**806**. If the system control unit **108** determines that the acquired lens information does not include the field angle information (NO in step S**805**), the processing proceeds to step S**807**. In step S**806**, the system control unit **108** generates second horizontal field angle information and second vertical field angle information based on the first horizontal field angle information, the first vertical field angle information, and the lens information acquired in step S**804** and the

imaging conditions determined in step S**801**. The processing then proceeds to step S**809**. Formulas (3) and (4) can be obtained when the first horizontal field angle is represented by AOV1_H, the first vertical field angle is represented by AOV1_V, the second horizontal field angle is represented by AOV2_H, the second vertical field angle is represented by AOV2_V, the diagonal line of the 35 mm fullsize sensor is represented by 35 mmFullSize_HV, and the diagonal line of the super 35 mm sensor is represented by Super35 mm_HV. In step S**807**, the system control unit **108** acquires focal length information in place of the first horizontal field angle information and the first vertical field angle information. The focal length information can be acquired from the lens information acquisition unit **115**, or can be acquired using the parameter storage unit **116** or the operation unit **114**, or can be acquired via the external I/F unit **113**. In step S**808**, the system control unit **108** generates third horizontal field angle information and third vertical field angle information based on the focal length information acquired in place of the first horizontal field angle information and the first vertical field angle information acquired in step S**807**, the imaging range information determined in step S**801**, and the like. Then, the processing proceeds to step S**809**. In the imaging range information determined in step S**801**, the following formulas (5) and (6) can be obtained when the horizontal imaging range is represented by Super35 mm_H, the vertical imaging range is represented by Super35 mm_V, the focal length is represented by FL, the third horizontal field angle is represented by AOV3_H, and the third vertical field angle is represented by AOV3_V.

(56) $AOV3\_H = 2 \cdot \tan^{-1}\left(\frac{Super35mm\_H}{FL}\right)$ (5) $AOV3\_V = 2 \cdot \tan^{-1}\left(\frac{Super35mm\_V}{FL}\right)$ (6)

(57) In step S**809**, the system control unit **108** selects the second horizontal field angle and second vertical field angle information or the third horizontal field angle and third vertical field angle information as the CG combining field angle information, and generates image capturing information, such as the second subject distance information as CG combining subject distance information, horizontal image field information, and vertical image field information.

(58) In step S**810**, the system control unit **108** associates the image capturing information generated in step S**809** and the pieces of lens information acquired in step S**804** with frames obtained during image capturing, and stores the image capturing information and the pieces of information associated with the frames in the system memory **109**. For example, the image capturing information and the pieces of information can be associated with frames by synthesizing the frames with the various information using time code. If video image recording is started in step S**803**, the recording can be completed in step S**811**.

(59) As described above, even when the user cannot recognize information for CG combining processing, field angle information for CG combining processing is generated based on imaging information, thereby making it possible to provide metadata for the user to accurately combine CG with a video image.

Fourth Exemplary Embodiment

(60) While the first exemplary embodiment described above illustrates an example where the subject distance information is used to generate metadata for CG combining processing in the image capturing apparatus **100**, a fourth exemplary embodiment illustrates an example where information about image fields is generated as metadata for CG combining processing in the image capturing apparatus **100**. An example of an imaging information generation method will be described with reference to the block diagram illustrated in FIG. **1** and a flowchart illustrated in FIG. **9**. The processing illustrated in FIG. **9** is repeatedly executed at an arbitrary predetermined cycle, such as an image capturing cycle. In the present exemplary embodiment, the image capturing cycle is used for the predetermined cycle.

(61) In step S**901**, the system control unit **108** determines imaging conditions and sets an exposure area for the image sensor **105**. In the present exemplary embodiment, a super 35 mm area is set in a 35 mm fullsize sensor, but instead any other imaging range can be set.

(62) In step S**902**, the system control unit **108** sets the imaging position, the focal length, the focus

position, and the like at initial positions. These settings can be made using the focus ring or the zoom ring mounted on the lens **101**, or through communication from the operation unit **114** to the lens **101**.

(63) In step S**903**, the system control unit **108** can start video image recording, or video image delivering via the external I/F unit **113** without recording the video image. In this case, the focal length, the focus position, and the like can be changed at any time.

(64) In step S**904**, the system control unit **108** acquires lens information, including focal length information, first subject distance information, and field angle information (first horizontal field angle and first vertical field angle), from the lens information acquisition unit **115**.

(65) In step S**905**, the system control unit **108** determines whether the lens information acquired in step S**904** includes the entrance pupil position and field angle information. If the system control unit **108** determines that the acquired lens information includes the entrance pupil position and field angle information (YES in step S**905**), the processing proceeds to step S**906**. If the system control unit **108** determines that the acquired lens information does not include the entrance pupil position and field angle information (NO in step S**905**), the processing proceeds to step S**908**.

(66) In step S**906**, the system control unit **108** generates the second subject distance information based on the first subject distance information and the lens information acquired in step S**904** and the imaging conditions determined in step S**901**. Then, the processing proceeds to step S**907**. Formula (1) can be obtained when the first subject distance information is represented by OD1, the entrance pupil position is represented by EP, and the second subject distance information is represented by OD2.

(67) In step S**907**, the system control unit **108** generates second horizontal field angle information and second vertical field angle information based on the second subject distance information generated in step S**906**, the first horizontal field angle information, the first vertical field angle information, and the lens information acquired in step S**904**, and the imaging conditions determined in step S**901**, and thereafter generates first image field information (first horizontal image field information and first vertical image field information). Then, the processing proceeds to step S**912**. Formulas (3) and (4) are obtained when the first horizontal field angle is represented by AOV1_H, the first vertical field angle is represented by AOV1_V, the second horizontal field angle is represented by AOV2_H, the second vertical field angle is represented by AOV2_V, the diagonal line of the 35 mm fullsize sensor is represented by 35 mmFullSize_HV, and the diagonal line of the super 35 mm sensor is represented by Super35 mm_HV.

(68) When the first horizontal image field is represented by FOV1_H and the first vertical image field is represented by FOV1_V, the following formulas (7) and (8) can be obtained.

(69) $$\mathrm{FOV1\_H} = \mathrm{OD2} \text{ .Math. } 2 \text{ .Math. } \tan\left(\frac{\mathrm{AOV2\_H}}{2}\right) \quad (7)$$

$$\mathrm{FOV1\_V} = \mathrm{OD2} \text{ .Math. } 2 \text{ .Math. } \tan\left(\frac{\mathrm{AOV2\_V}}{2}\right) \quad (8)$$

(70) In step S**908**, the system control unit **108** acquires a coefficient in place of the entrance pupil position. The coefficient can be acquired from the parameter storage unit **116**, or can be acquired using the operation unit **114**, or can be acquired via the external I/F unit **113**. In step S**909**, the system control unit **108** generates the third subject distance information based on the coefficient acquired in place of the entrance pupil position in step S**908**, the first subject distance information acquired in step S**904**, and the imaging conditions determined in step S**901**. Then, the processing proceeds to step S**910**. Formula (2) is obtained when the first subject distance information is represented by OD1, the coefficient acquired in place of the entrance pupil position is represented by K1, and the third subject distance information is represented by OD3.

(71) In step S**910**, the system control unit **108** acquires focal length information in place of the first horizontal field angle information and the first vertical field angle information. The focal length information can be acquired from the lens information acquisition unit **115**, or can be acquired using the parameter storage unit **116** or the operation unit **114**, or can be acquired via the external

I/F unit **113**.

(72) In step S**911**, the system control unit **108** generates third field angle information (third horizontal field angle information and third vertical field angle information) based on the focal length information acquired in place of the first horizontal field angle information and the first vertical field angle information acquired in step S**907**, the focal length information acquired in step S**904**, and the imaging conditions determined in step S**901**. Thereafter, the system control unit **108** generates second field angle information (second horizontal field angle information and second vertical field angle information), and then generates second horizontal image field information and second vertical image field information. Then, the processing proceeds to step S**912**. Formulas (5) and (6) are obtained when the horizontal imaging range is represented by Super35 mm_H, the vertical imaging range is represented by Super35 mm_V, the focal length is represented by FL, the third horizontal field angle is represented by AOV3_H, and the third vertical field angle is represented by AOV3_V, in the imaging conditions determined in step S**901**. Further, when the second horizontal image field is represented by FOV2_H and the second vertical image field is represented by FOV2_V, the following formulas (9) and (10) can be obtained.

(73) $$FOV2\_H = OD3 \times 2 \times \tan\left(\frac{AOV3\_H}{2}\right) \quad (9)$$

$$FOV2\_V = OD3 \times 2 \times \tan\left(\frac{AOV3\_V}{2}\right) \quad (10)$$

(74) In step S**912**, the system control unit **108** associates the imaging information generated in step S**911** and the pieces of information acquired in step S**904** with frames obtained during image capturing, and stores the imaging information and the pieces of information associated with the frames in the system memory **109**. For example, the imaging information and the pieces of information can be associated with the frames by synchronizing the frames with the various information using time code. If video image recording is started in step S**903**, the recording can be completed in step S**913**.

(75) As described above, image field information for CG combining processing is generated based on imaging information even when the user cannot recognize information for CG combining processing, thereby making it possible to provide metadata for the user to accurately combine CG with a video image.

(76) While the present disclosure is described in detail above based on the exemplary embodiments, the present disclosure is not limited to these specific exemplary embodiments. Various modifications can be made without departing from the scope of the disclosure. Further, some of the exemplary embodiments described above can also be extracted and combined as needed. In each exemplary embodiment, the CG combining subject distance information, the CG combining field angle information, the CG combining image field information are calculated by various methods, but instead can be obtained by any other method.

(77) According to an aspect of the present disclosure, it is possible to implement appropriate CG combining processing even when the user can recognize only a distance from an imaging plane to a subject as subject distance information.

Other Embodiments

(78) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described

embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(79) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(80) This application claims the benefit of Japanese Patent Application No. 2022-031995, filed Mar. 2, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image processing apparatus comprising: one or more memories storing instructions; and one or more processors that, upon execution of the instructions, is configured to operate as: an acquisition unit configured to acquire an image and first subject distance information, corresponding to the image and indicating a distance from an image sensor to a subject, which are generated by an image capturing apparatus including the image sensor configured to capture a light beam from a subject through an optical lens; a generation unit configured to generate second subject distance information indicating a distance from the image capturing apparatus to the subject based on information corresponding to an entrance pupil position of the optical lens used to capture the image, a focus position, and the first subject distance information; and an output unit configured to output, as metadata, the second subject distance information generated by the generation unit in association with the image.
2. The image processing apparatus according to claim 1, wherein the information corresponding to the entrance pupil position of the optical lens is information indicating an entrance pupil distance of the optical lens.
3. The image processing apparatus according to claim 1, wherein the information corresponding to the entrance pupil position of the optical lens is information stored based on an actual measured value in each state of the optical lens.
4. The image processing apparatus according to claim 1, wherein the generation unit generates image field information, based on information corresponding to a field angle of each of the optical lens and the image sensor used to capture the image and information about a size of the image sensor.
5. The image processing apparatus according to claim 4, wherein the output unit configured to output, as metadata, the image field information generated by the generation unit in association with the image.
6. The image processing apparatus according to claim 4, wherein the information corresponding to the field angle is generated based on a focal length for capturing the image.
7. The image processing apparatus according to claim 1, further comprising a computer graphics (CG) combining unit configured to combine one of the image and a video image with CG using the metadata.
8. The image processing apparatus according to claim 1, wherein the output unit synchronizes the image and the metadata and outputs the image and the metadata in real time.
9. A control method for an image processing apparatus, the control method comprising: acquiring an image and first subject distance information, corresponding to the image and indicating a

distance from an image sensor to a subject, which are generated by an image capturing apparatus including the image sensor configured to capture a light beam from a subject through an optical lens; generating second subject distance information indicating a distance from the image capturing apparatus to the subject based on information corresponding to an entrance pupil position of the optical lens used to capture the image, a focus position, and the first subject distance information; and outputting, as metadata, the second subject distance information generated by the generation unit in association with the image.

10. The control method according to claim 9, wherein the information corresponding to the entrance pupil position of the optical lens is information indicating an entrance pupil distance of the optical lens.

11. The control method according to claim 10, wherein in the outputting, as metadata, the image field information in association with the image.

12. The control method according to claim 9, wherein the information corresponding to the entrance pupil position of the optical lens is information stored based on an actual measured value in each state of the optical lens.

13. The control method according to claim 9, wherein image field information is generated, based on information corresponding to a field angle of each of the optical lens and the image sensor used to capture the image and information about a size of the image sensor.

14. The control method according to claim 13, wherein the information corresponding to the field angle is generated based on a focal length for capturing the image.

15. The control method according to claim 9, further comprising combining one of the image and a video image with CG using the metadata.

16. The control method according to claim 9, wherein the image and the metadata are synchronized and output the image and the metadata in real time.